Exp 24 MATLAB CODE

```
clc;
clear;
close all;

% Parameters
N = 10000;
Fs = 100000;
fc = 10000;
SNR = 10;

% Generate Random Bits
data = randi([0 1],1,N);

% BPSK Modulation
symbols = 2*data - 1;

% Passband Transmission
t = (0:N-1)/Fs;
tx = symbols .* cos(2*pi*fc*t);

% Multipath Fading Channel
h = [1 0.5 0.3];
faded_signal = filter(h,1,tx);

% Add AWGN Noise
signal_power = mean(faded_signal.^2);
noise_power = signal_power/(10^(SNR/10));
```

```
rx = faded_signal + sqrt(noise_power)*randn(size(faded_signal));

% Coherent Demodulation
demod = rx .* cos(2*pi*fc*t);

% Low-Pass Filtering
[b,a] = butter(5,2000/(Fs/2),'low');
baseband = filtfilt(b,a,demod);

% Remove Initial Transient
start = 100;
detected = baseband(start:end) > 0;
original = data(start:end);

L = min(length(detected),length(original));
detected = detected(1:L);
original = original(1:L);

% BER Calculation
errors = sum(original ~= detected);
BER = errors/L;

fprintf('SNR = %d dB\n',SNR);
fprintf('Total Bits = %d\n',L);
fprintf('Bit Errors = %d\n',errors);
fprintf('Bit Error Rate (BER) = %.6f\n',BER);

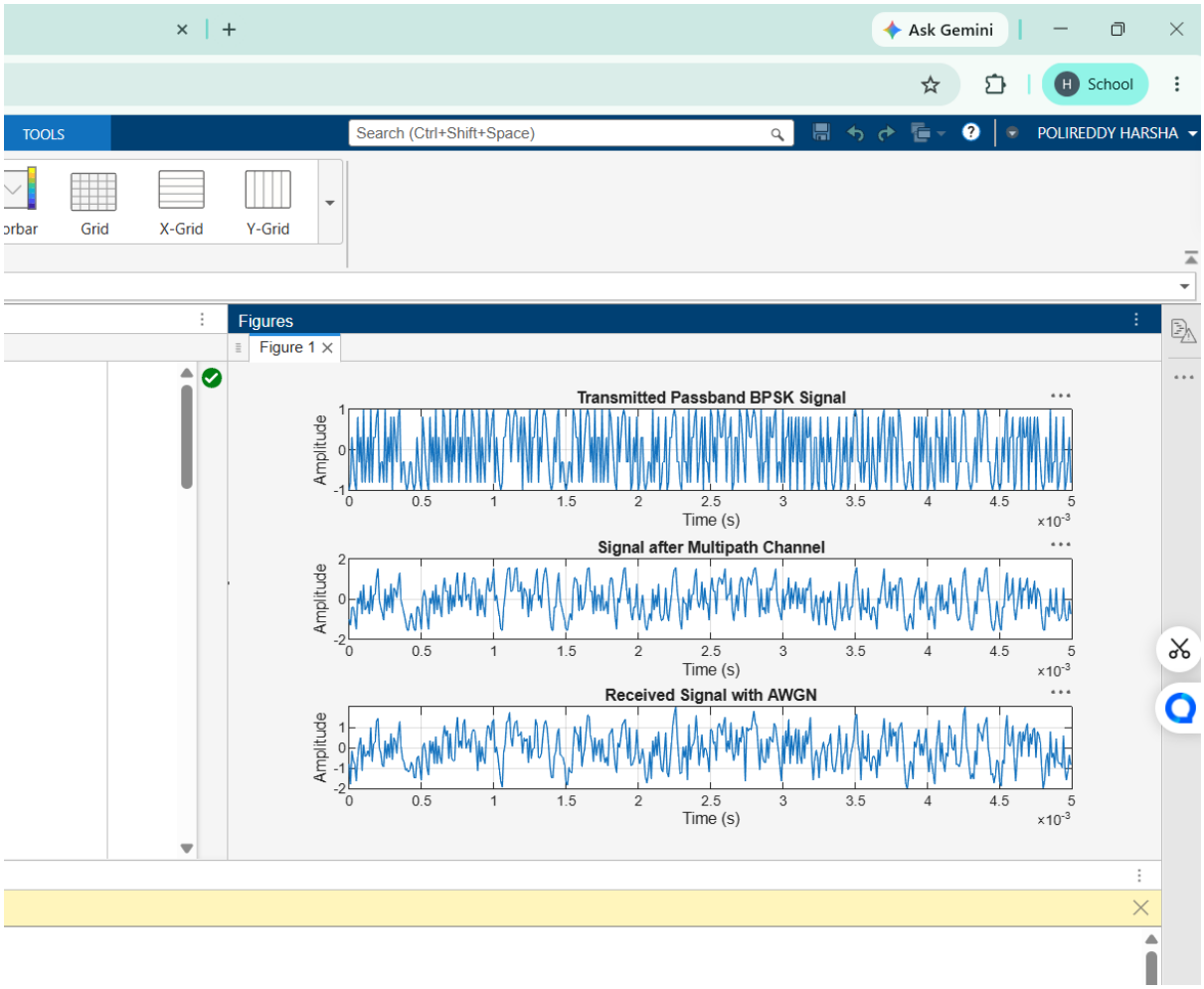
% Plot Results
```

```
figure;

subplot(3,1,1);
plot(t(1:500),tx(1:500));
title('Transmitted Passband BPSK Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
plot(t(1:500),faded_signal(1:500));
title('Signal after Multipath Channel');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,3);
plot(t(1:500),rx(1:500));
title('Received Signal with AWGN');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```


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Figures
Figure 1
Transmitted Passband BPSK Signal
Amplitude
Time (s)
×10-3
Signal after Multipath Channel
Amplitude
Time (s)
×10-3
Received Signal with AWGN
Amplitude
Time (s)
×10-3